ANGELA ALTAMIRANO

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https://angelaalta.github.io/engineeringportfolio/index.html

EDUCATION

Cornell University, College of Engineering, Ithaca, NY — Expected December 2025
Master of Engineering in Mechanical Engineering

Cornell University, College of Engineering, Ithaca, NY — May 2025
Bachelor of Science in Biomedical Engineering, Cum Laude, GPA: 3.6
Awards: Ryan's Scholar, Louis Stokes Alliance for Minority Participation (LSAMP) Scholar, Dean's List (4 Semesters)

SPECIALIZED SKILLS

Programming Languages: Matlab, Python, RStudio, Julia

Modeling & Simulation: Fusion 360, SolidWorks, ANSYS Fluent, COMSOL, AFNI, ImageJ, Microsoft Office Suite

Laboratory & Manufacturing: Cell Culture, Southern Blot Analysis with PCR, Signal Measuring with Oscilloscope, Soldering and Circuitry Design, Arduino, 3D Printing, Manual Mill, Manual Lathe, GD&T, DFMA, DFMEA

RELEVANT EXPERIENCE

Senior Capstone Project: EasiStitch, Ithaca, NY, Mechanical Lead — August 2024 - May 2025
- Designed and built a gear-based mechanical system for EasiStitch, an automated suturing device, producing iterative SolidWorks models, photorealistic renders, and a functional "works-like" prototype via 3D printing
- Translated cross-disciplinary stakeholder input from emergency, surgical, and pediatric providers in varied economic settings into design adaptations that enhanced device functionality, ergonomics, and usability

Jacobs Institute, Buffalo, NY, Engineering Intern — June - August 2024
- Collaborated with the R&D team and shadowed operating room clinicians to improve 3D-printed cardiovascular model haptics and design, optimizing vascular structures for realistic simulation of in vivo vessel behavior
- Conducted biomechanical characterization of vascular models under simulated use conditions, performing tensile testing on SolidWorks-designed samples using an Instron-68SC to assess model robustness

Cornell DEBUT, Ithaca, NY, Project Manager — August 2023 - May 2024
- Led a team of 7 R&D Analysts to develop *SteadyStride: A Self-Stabilizing Cane for Parkinson's Disease*, integrating 3D-printed components designed in SolidWorks, MATLAB-based modeling and simulation of device dynamics, and iterative prototyping to optimize tremor mitigation
- Served as Principal Investigator for an IRB-approved human subject study at SUNY Cortland, overseeing experimental protocol, accelerometer-based motion data acquisition, and statistical analysis to quantify device performance
- Awarded First Place at Medtronic/BMES Student Design Competition (October 2024)

Weill Cornell Biomedical Engineering Immersion Program, New York, NY, Scholar — January 2024
- Participated in a two-week clinical immersion program with rotations across Emergency Medicine, Hospital Medicine, Neurosurgery, Anesthesiology, Plastic Surgery, and Radiology, observing workflows and clinical decision-making

Cornell Engineering: Biomedical Circuits, Signals and Systems, Ithaca, NY, Tutor — August - December 2024
- Provided individualized tutoring for BME 3030, supporting student mastery of topics such as circuit analysis, signal processing, and system modeling in biomedical applications

ADDITIONAL EXPERIENCE

Cornell Affect and Cognition Lab, Ithaca, NY, Research Assistant — August 2023 - May 2025
- Facilitated MRI scans for Parkinson's Disease patients and utilized AFNI for image analysis

Placental Membrane Characterization, Ithaca, NY, Research Assistant — August - December 2024
- Worked within a team of 6 and partnered with MTF Biologics to test the mechanical and material properties of AmnioBand® Membrane (Allograft Placental Matrix) through tensile testing, strain mapping, and three-point bending
- Used ANSYS to create a 3D model with biological inputs to validate and compare the experimental three-point bending results